PCI-SIG ENGINEERING CHANGE NOTICE

TITLE:	Revised _DSM for Cache Locality TPH Features
DATE:	Draft version -- September 21, 2020
AFFECTED DOCUMENT:	PCI Firmware Specification 3.3
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Part I

1. Summary of the Functional Changes

This ECN extends the functionality provided by the TPH Features _DSM introduced in Revision 3.3. of the PCI Firmware Specification.

2. Benefits as a Result of the Changes

The added functionality provides support for standard use-cases, using well-defined input and output parameters.

3. Assessment of the Impact

The added functionality is applicable to the PCI Bus Driver and the ACPI interface. Additionally, device drivers utilizing system-specific ST values would be updated to support a new PCI bus driver interface (in an OS-implementation-specific manner).

4. Analysis of the Hardware Implications

There is no required impact to HW to support this ECN. This is an optional capability to convey the platform's support of the specified capabilities over the PCIe interface, using the optional TLP Processing Hint (TPH) Extended Capability.

5. Analysis of the Software Implications

Operating system software can use this interface to discover platform-defined Steering Tag values to support the specified functionality. The PCI bus driver coordinates usage of a declared ST value to client device drivers, and the device drivers configure their hardware to enable the ST for the desired feature.

6. Analysis of the C&I Test Implications

No impact; enables the specified standard features.

Part II

Detailed Description of the change

Markup Conventions:

Black:	*Existing text from PCI Firmware Specification, Revision 3.3 (https://members.pcisig.com/wg/Firmware/document/13944)*
~~Red Strikethrough~~:	*Deleted text*
Blue Underlined:	*New text*

Revise Section 4.6 as follows:

4.6. _DSM Definitions for PCI

...

The Current Revision ID value for this revision of this specification is ~~6~~7.

4.6.15. _DSM to Query Cache Locality TPH Features

The Query Cache Locality TPH Features function provides system-specific Steering Tag (ST) values to support Cache Locality features that allow communication of TLP Processing Hints between PCI Express Endpoints and their corresponding Root Complex.

The returned ST values may be used by Endpoints that implement the TPH Requester Extended Capability structure, and support Interrupt Vector Mode or Device Specific ST Mode of operation. The values returned by this _DSM function also apply to Root Complex Integrated Endpoints.

For more details, refer to the **TLP Processing Hints (TPH)** section of the PCI Express Base Specification.

Location:

Support for this function is only applicable for a _DSM Method within the scope of a PCI Express Host Bridge.

Arguments:

Arg0: UUID: E5C937D0-3553-4d7a-9117-EA4D19C3434D

Arg1: Revision ID: lowest valid Revision ID value: 7

Arg2: Function Index: 0Fh.

Arg3: Package:

```
Package {
        FeatureID                   // DWORD (32 bits)
        FeatureArgument1            // DWORD (32 bits)
        FeatureArgument2            // QWORD (64 bits)
}
```

Refer to Table 4.x for **FeatureID** definitions.

- **For FeatureID == 0 (Processor Cache Steering Tags)**

FeatureArgument1: DWORD:

Bits	Description
[31:0] – Target UID	If the target is a processor, then this field represents the ACPI Processor UID of the processor as specified in the MADT. If the target is a processor container, then this field represents the ACPI Processor UID of the processor container as specified in the PPTT. Refer to the Target Type field of FeatureArgument2 to determine the target type.

FeatureArgument2: QWORD:

Bits	Description
[1:0] – Processing Hint (PH)	The PH bits signify that the requested Steering Tags are intended to support the usage model that corresponds to the PH encodings defined in the PCI Express Base Specification.
[2] – Target Type	Specifies whether the Target UID value specified in FeatureArgument1 refers to a processor or to a processor container. 0 – The Target UID specified in FeatureArgument1 refers to a processor. 1 – The Target UID specified in FeatureArgument1 refers to a processor container.
[3] – Cache Reference Valid	If set, this indicates that the Cache Reference field is valid.
[31:4]	Reserved.
[63:32] – Cache Reference	The Cache ID of the specific cache. The Cache ID must match the ACPI Cache ID field in the PPTT Type 1 structure (Cache Structure) that describes this cache.

Return Value: QWORD: Refer to Table 4.y.

- **For any other FeatureID:**

  FeatureArgument1: DWORD: Reserved.

  FeatureArgument2: QWORD: Reserved.

  Return Value: QWORD: 0 (Not supported).

Table 4-x: FeatureID Definitions

FeatureID	**Description**
0 - Processor Cache Steering Tags	Defines Steering Tags values that may be used to steer DMA requests to a specific cache or a cache that is as close as possible to a target processor.
Any other value	Reserved.

Table 4-y: Return Value

Bits	**Description**
[0] – VolatileMemory ST Valid	If zero, the VolatileMemory ST field is ignored by OSPM.
[1] – VolatileMemory ExtendedST Valid	If zero, the VolatileMemory ExtendedST field is ignored by OSPM.
[2] – VolatileMemory PH Ignore	If set, the PH argument was ignored, and that PH value will be ignored if supplied as a TLP Hint. If clear, the PH argument was considered, and the PH value should be supplied as a TLP Hint.
[7:4] - Reserved	Reserved. Must be 0.
[15:8] – VolatileMemory ST	8-bit ST value used when addressing Volatile Memory.
[31:16] – VolatileMemory ExtendedST	16-bit ST value that may be used for more precise steering. **Note:** Both the 8-bit and the 16-bit Tags may be Valid, and the lower 8 bits of the 16-bit Tag may be different than the 8-bit Tag.
[32] – PersistentMemory ST	If zero, the PersistentMemory ST field is ignored by OSPM.
[33] – PersistentMemory ExtendedST Valid	If zero, the PersistentMemory ExtendedST field is ignored by OSPM.
[34] – PersistentMemory PH Ignore	If set, the PH argument was ignored, and that PH value will be ignored if supplied as a TLP Hint. If clear, the PH argument was considered, and the PH value should be supplied as a TLP Hint.
[39:35] - Reserved	Reserved.
[47:40] – PersistentMemory ST	8-bit ST value used when addressing Persistent Memory.
[63:48] – PersistentMemory ExtendedST	16-bit ST value that may be used for more precise steering. **NOTE:** Both the 8-bit and the 16-bit Tags may be Valid, and the lower 8 bits of the 16-bit Tag may be different than the 8-bit Tag.

Note: VolatileMemory refers to a region of memory that operates as *EfiConventionalMemory*, as defined by the UEFI Spec. Similarly, **PersistentMemory** refers to a region of memory that operates as *EfiPersistentMemory*, as defined by the UEFI Spec.

IMPLEMENTATION NOTE

Disabling TPH Steering Feature

If VolatileMemory ST Valid == 0, VolatileMemory Extended ST Valid == 0, and VolatileMemory PH Ignore == 1 for all requested PH encodings, then the TPH Steering feature should be disabled when addressing Volatile Memory.

Similarly, if PersistentMemory ST Valid == 0, PersistentMemory Extended ST Valid == 0, and PersistentMemory PH Ignore == 1 for all requested PH encodings, then the TPH Steering feature should be disabled when addressing Persistent Memory.